

COURSE: CHE 312 – Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Chicago, Illinois 60607

Overview-Objectives: Heat and mass transport phenomena. Heat conduction, convection and radiation. Heat exchanger design. Diffusion. Mass transfer coefficients.

The broad objectives of this course are:

- (a) To develop clear understanding of the basic mechanisms of heat and mass transfer in physical processes encountered in daily living experiences and in chemical engineering practice; and
- (b) To develop the skills to describe heat and mass processes in terms of mathematical models, make appropriate assumptions to simplify the models, and perform calculations to obtain qualitative and/or quantitative information which can lead to better design or better understanding of the processes.

Instructor: Professor Christos G. Takoudis, SEO 207, Phone #: (312) 355-0859, Login: takoudis@uic.edu. Office Hours: W 1:00-2:00 pm or anytime by appointment in SEO 207

Lecture: Class meets on MWF at 12:00 pm, in Thomas Beckham Hall 180F.

Teaching Assistants: Srishyam Raghavan, Phone #: (779) 3330037, Login: sragha20@uic.edu (office hours: T 11:30 am – 1:00 pm, R 4:30 – 5:30 pm, and F 4:00 – 5:30 pm in EIB 228); Rajan Bhawnani Phone #: (848) 2528207, Login: rbhawn2@uic.edu (office hours: M 4:00 – 5:00 pm, and W 3:00 – 4:00 pm in EIB 271).

Outline of Topics:

Introduction

Introduction to Conduction

One-Dimensional Steady-State Conduction

Two-Dimensional Steady-State Conduction

Transient Conduction

EXAM I

Introduction to Convection
 Heat Transfer in External Flows
 Heat Transfer in Internal Flows
 Free Convection
 Heat Exchangers

EXAM II

Radiation Heat Transfer
 Radiation Exchange Between Surfaces
 Diffusion Mass Transfer
 Diffusion – Homogeneous Chemical Reaction Systems

EXAM III

Diffusion – Heterogeneous Chemical Reaction Systems
 Transient Diffusion and Applications
 Diffusion and Chemical Reaction Inside a Porous Catalyst
 Mass Transfer Design Calculations
 Multimode Heat and Mass Transfer

EXAM IV

Prerequisite: CHE 311

Homework: Approximately 12 homework assignments.

Examinations: Four one-hour exams. Only one paper card 3 x 5 (on which you may write equations, etc.) per student is allowed in the exams. The approximate dates of the examinations are:

Exam I (One-Hour): After about 4 weeks of classes
Exam II (One-Hour): After about 8 weeks of classes
Exam III (One-Hour): After about 12 weeks of classes
Exam IV (One-Hour): After about 15 weeks of classes

The exact date of each exam will be announced at least one week before the exam.

There will be unannounced quizzes based on lecture material and the reading assignments. Each quiz will last about 15 minutes. *There will be no early or make-up exams or quizzes.*

Grade: The final score for the course will be computed as follows:

Homework Assignments:* 15 %

(*A minimum of 2/3 completed HW Assignments is required for a passing grade. For changes to HW grades, please see your TA within 2 weeks of posting of grades. In case of illness

please let your TA know as soon as possible - failure to do so will be considered as late submission and marks will be deducted.)

<i>Unannounced Quizzes</i>	17 %
<i>Exam I:</i>	17 %
<i>Exam II:</i>	17 %
<i>Exam III:</i>	17 %
<i>Exam IV:</i>	17 %

Textbook:

Theodore L. Bergman, Adrienne S. Lavine, *Fundamentals of HEAT and MASS Transfer*, Wiley, Eighth Edition, 2019

Other Suggested References:

Bird R., Stewart W., and Lightfoot E., *Transport Phenomena*, John Wiley & Sons, 2002 (2nd Edition).

Additional Notes